Mizan Tepi UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATICS (CCI)

DEPARTMENT OF INFORMATION TECHNOLOGY

COURSE TITLE FUNDAMENTAL OF DATABASE SYSTEMS PROJECT

TITLE: MTU CLINICAL INFORMATION MANAGEMENT SYSTEM

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ACRONYM

ER – Entity Relationship **ERD – Entity Relationship Diagram**

MTU — University **NF – Normal Form**

1NF – First Normal Form **2NF – Second Normal Form**

SQL – Structural Query Language

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Chapter 1

1.1 INTRODUCTION

The current system used at the Mizan Tepi University Health Centre is paper-basedand operates manually. Patients go through a card registration process where theirinformation is recorded on paper. This data is then taken to the doctor for examination. If any laboratory tests are required, the doctor writes a request onpaper, which the patient takes to the laboratory. Once thetests are completed, theresults are documented on paper and returned to thedoctor. If medication is needed,the doctor writes a prescription on paper, whichthe patient takes to the pharmacy.Unfortunately, this process is slow, time-consuming, and often leads to delays andunnecessary labor.

To address these challenges, Mizan Tepi University plans to introduce a new digitalsystem that will interconnect the campuses and replace the manual process. Theproposed system aims to achieve better control over patient information andmedical records, ultimately reducing both time and human labor. Doctors, laboratory technicians, pharmacists, and card receptionists will each havetheir own accountsto manage patient information and requests. Additionally, thesystem will maintainrecords of students who leave the university, allowing theirmedical histories to beretrieved when they return. Overall, this digital systempromises to create a fasterand more efficient work environment for everyoneinvolved.

1.2 Background

Mizan-Tepi University Light of the Green Valley

Mizan-Tepi University (MTU) in Ethiopia started as a health sciences institute in 1983, grew into a full university in 2006, and is known as an innovative, community-focused institution in the Southwest region with campuses in Mizan-Aman and Tepi, offering programs in health, tech, business, education, and sciences, focusing on skilled manpower and regional development with its Teaching Hospital.

clinic is currently in the process of being upgraded to the level of a health centre with the completion of a new building and acquisition of complete and up-to-date healthcare equipment and facilities. The clinic's workflows are done within twenty-four hours, ensuring that patients receive timely and efficient services.

The health centre offers a wide range of basic services to its patients. These include emergency services, outpatient services, inpatient services, maternal and child health, family planning, delivery services, general laboratory services, pharmacy services, paediatric, internal medicine, enabling services, and much more. The health centre operates with the goal of offering affordable and quality healthcare to all students, and it has made significant strides in achieving this objective. In conclusion, Mizan tepi University is not only a reputable educational institution but also a provider of quality healthcare services. Its clinic is continuously being improved and expanded, enabling it to meet the ever-growing healthcare needs of its students. Sure, here's the revised literature review focused on clinical management systems:

1.3 Literature Review

Clinical management systems are pivotal for healthcare organizations, facilitating the efficient management of patient care, administrative tasks, and resource allocation. This literature review delves into the advantages of employing a clinical management system within healthcare settings.

One of the paramount advantages of a clinical management system is centralized data storage. By consolidating patient information, treatment plans, and medical records into a single, integrated platform, clinical management systems enhance data accuracy and streamline access for healthcare professionals. This centralized repository not only reduces redundancies in data entry but also ensures that critical patient information is readily accessible, promoting informed decision-making and enhancing patient care outcomes. Moreover, robust security measures inherent in these systems safeguard patient confidentiality and mitigate the risk of unauthorized access or data breaches.

Furthermore, clinical management systems offer unparalleled capabilities in

reporting and analytics. By harnessing real-time data insights, healthcare administrators can gain valuable perspectives on patient demographics, treatment outcomes, and resource utilization. Armed with this information, healthcare leaders can identify trends, assess performance metrics, and refine clinical protocols to optimize patient care delivery. Additionally, customizable reporting features empower healthcare managers to generate tailored dashboards and metrics, facilitating ongoing monitoring and evaluation of clinical operations. Automation is another key benefit of clinical management systems, streamlining routine administrative tasks and enhancing operational efficiency. Functions such as appointment scheduling, billing, and inventory management can be automated, freeing up healthcare staff to focus on patient care and clinical decision-making. For instance, automated appointment reminders and patient notifications minimize no-show rates and enhance appointment adherence, while automated billing processes accelerate revenue cycle management and reduce billing errors. Moreover, self-service portals empower patients to actively engage in their healthcare journey, enabling them to schedule appointments, access medical records, and communicate with healthcare providers seamlessly. In summary, clinical management systems offer a multifaceted approach to healthcare administration, revolutionizing patient care delivery through centralized data storage, robust analytics, and streamlined automation. By embracing these systems, healthcare organizations can optimize operational workflows, enhance patient outcomes, and drive continuous improvements in clinical practice.

1.4 Statement of Problem

The way clinical data is managed at MTU University has many problems that make it hard to work efficiently. One big issue is how much time and effort it takes to use the current system. Doing things manually with paper records makes it tough to get and understand patient information quickly. This causes delays in getting important patient details when needed. In today's world where the internet and digital tools are common, relying on paper records is not a good idea. It's clear that continuing to use paper records can lead to many problems that affect how well we can access and use the data.

Data Entry Errors: Manual data entry is prone to mistakes. Even a small error can have significant consequences, especially in healthcare. These errors can impact patient care decisions and lead to serious health complications.

Time-Consuming Processes: Manual data handling is time-consuming, which can affect overall efficiency. Healthcare workers spend valuable time entering data instead of focusing on patient care.

Dependency on Professionals: Relying solely on manual data handling creates a heavy dependency on individual professionals. If they are unavailable or make errors, it can disrupt operations.

Limited Scalability: As the volume of healthcare data grows, manual processes become less scalable. Handling large amounts of data manually can overwhelm staff and lead to inefficiencies and Paper-based records are susceptible to damage, loss, or misplacement. Losing critical patient information can have serious consequences.

Difficulty in Data Analysis: Manual data handling makes it challenging to analyze trends, patterns, and outcomes. Data-driven decision-making becomes cumbersome

1.5 Objectives

1.5.1 General objective

The main goal of this project is to convert the manual data handling method of clinic to the computerized form.

1.5.2 Specific objective

- ★ To record the patient and employee information
- ★ To retrieve and update data easily
- ★ To manage patient and employee
- ★ To reduce the redundancy of the data
- ★ The reduce the use of the paper work
- ★ To reduce the cost and time

1.6 Scope

The scope of the project covers the computer-based information system for a hospital used by System manager (admin), patient, card receptionist, Doctor, laboratory Pharmacy.

- ★ The system will record the patient and employee information.
- ★ The system will provide easy access of data.
- ★ The system will make it easier to contact between employees.
- ★ The system can manage all employees and patient information.
- ★ The system can edit patient and employees information.
- ★ The system can modify patient and employees data.

1.7 Limitation and constraints

1.7.1Limitation

The system we developed is not web-based: This means that our clinical database management system cannot be accessed through a web browser. Users cannot interact with the system via the internet, limiting accessibility and remote usage. As a result, users may be restricted to accessing the system only from specific locations or devices where the software is installed.

The system we developed did not utilize various technologies, including cloud-based platforms: Our clinical database management system does not leverage modern technologies such as cloud computing. Cloud-based platforms offer benefits such as scalability, flexibility, and enhanced data security. However, our system lacks these advantages, potentially limiting its ability to adapt to changing needs and efficiently manage large volumes of data.

The university did not allocate a budget to cover the cost of hardware, software, and development: Due to budget constraints, we were unable to invest in the necessary hardware, software licenses, and development resources required to build a more comprehensive and robust clinical database management system. This limitation may have hindered the implementation of advanced features, scalability, and overall system performance.

1.7.2 Constraint

Lack of time to collect data: Due to time constraints, there may have been insufficient opportunity to gather all the necessary data required for the development or implementation of the clinical database management system. This limitation can result in incomplete or inaccurate information being incorporated into the system, potentially compromising its effectiveness and reliability.

Difficulty in getting information from stakeholders: Difficulty in obtaining relevant information from stakeholders, such as healthcare providers, administrators, or patients, can hinder the development and implementation process. Without access to key data and insights from stakeholders, it becomes challenging to understand user requirements, design a system that meets their needs, and ensure the system's alignment with organizational goals.

Lack of support from stakeholders: The absence of support or cooperation from stakeholders can impede progress and hinder the successful implementation of the clinical database management system. Without buy-in from key stakeholders, such as healthcare professionals, management, or IT personnel, there may be resistance to change, reluctance to adopt the new system, or insufficient resources allocated to support its implementation and maintenance.

1.8 Significant of the project

Time saving for both patients and employees: By implementing a streamlined clinical database management system, the project aims to reduce the time required for various tasks, such as appointment scheduling, data entry, and accessing patient records. This time-saving benefit not only improves efficiency within the clinic but also enhances the overall patient experience by minimizing waiting times and administrative delays.

Ease of managing and recording data: The project seeks to simplify the process of managing and recording clinical data by transitioning from manual paper-based methods to digital systems. Digital databases enable easier data entry, organization, and retrieval, reducing the likelihood of errors and ensuring that accurate and up-to-

date information is readily available for healthcare providers when needed.

Reduction in paper usage and wastage: Moving away from paper-based record-keeping towards digital databases helps minimize the use of paper and reduces associated wastage, such as printing unnecessary documents and storing physical records. This not only contributes to environmental sustainability but also saves costs associated with paper procurement, storage, and disposal.

Easy retrieval of pre-existing patient information: With a digital database system in place, healthcare providers can quickly retrieve pre-existing patient information, such as medical history, lab results, and treatment plans. This improves continuity of care, enables informed decision-making, and enhances patient safety by ensuring that healthcare providers have access to comprehensive patient records during consultations and procedures.

The significance of the project lies in its potential to optimize clinical processes, improve patient care outcomes, and enhance operational efficiency within the clinic, ultimately leading to better healthcare delivery and patient satisfaction.

1.9 Methodology

The methodology for transitioning the MTU clinical management system from amanual to a database system involves several key steps:

- ✓ Define Project Scope: This includes setting clear project objectives and identifying the necessary resources.
- ✓ Gather User Requirements: Conduct interviews with clinic staff and stakeholders to understand their needs and expectations. Obtain accurate and relevant information about the role and operations of MTU University's clinical management . We visited the MTU clinic manager, doctorand nurse in person and gathered insights through prepared questionnaires.

- ✓ Select Software Tools: Choose appropriate software tools for system design. For example, Wondershare EdrawMax and Microsoft SQL Server Studio will be utilized for creating the Entity-Relationship Diagram (ERD) and database tables, respectively.

- ✓ Design Database Schema: Carefully design the database schema by identifying entities, attributes, and relationships. Develop the ER diagram and relational schema using SQL

1.9.1 Observation

We observed the current working system of the clinic as a patient and caregiver. The system works in manual way of data handling so in order to get medical services first you have to take card from the patient registration office then your personal information will be written on the paper and the paper is stored in some file case then your file will be transferred either to the doctor, nurse or health officer.

After the file is sent to the medical assistant of the clinic you will get treatment. If the treatment needs more process you will go to the laboratory. After you finish your laboratory tests you will back to the first doctor or medical assistant you meet then he will give you some medications after that he will send you to the pharmacy for additional drugs. all of this process is done by paper work. And every time you go to the clinic your file will be searched by the card number they gave you last time of your treatment.

1.9.2 Interview

We used this method to gather information by questioning the concerned body at the clinic.The interview process was utilized as an effective means of gathering pertinent information by engaging in direct questioning with the relevant individuals or stakeholders at the clinic. This method facilitated a detailed exploration and understanding of the subject matter under investigation, enabling comprehensive data collection and insight into the clinic's operations or specific areas of interest.

1.10 Feasibility

The clinical management system is an essential component of healthcare facilities. It helps in managing patient information, medical records, and treatment plans. MTU University is planning to implement a clinical management system to enhance its healthcare services. The proposed clinical management system is viable and has the potential to improve the quality of healthcare services provided by MTU University. With proper planning, resources, and execution, the project can be successfully implemented, improving patient outcomes and increasing operational efficiency.

The proposed project, we have conducted a thorough analysis. The following factors were considered:

1. Technical feasibility: The proposed system can be developed using various technologies, including database management systems. MTU University hasthe technical expertise to develop the system or can outsource the development to a reputable third-party vendor.

2. Financial feasibility: The cost of developing and implementing the proposed system is a crucial factor.

3. Operational feasibility: The success of the proposed system depends on its operational feasibility. The system must be user-friendly, and staff must be trained to use it effectively. The existing processes should be streamlined, and the system should not disrupt existing healthcare operations.

4. Legal feasibility: The proposed system must comply with all legal and regulatory requirements. The system should ensure patient privacy and adhere to data protection regulation.

CHAPTER TW0

2.1 Body of work

This system is designed for MTU University Health Centre to replace their existing manual system (paper-based system) with a digital and more efficient one. The new system is intended to control patient information and medical records with the goal of reducing time and human labour. The proposed system of MTU University is to interconnect the campuses. Students of the university who graduate or leave the university and used to go to the health care to get treatments will leave at some time. The health centre has no idea if its patients have left or not, and the files of these patients will take a lot of space. Therefore, there is a need to record these students when they will leave or how long they will be in the university so that if they leave, it can be stored as deactivated because if there is any chance of them returning, their history can be taken from the deactivated files and reactivated as soon as they return.

The manager will have its own account which can be accessed by logging in as admin using a password and admin name. Doctors, laboratory technicians, pharmacists, and card receptionists will be able to have accounts that are created by the system. They can delete accounts and also manage employees. The accounts of doctors will have a doctor interface account which includes the doctor's name and password to log in to the account. They will be able to perform tasks such as sending laboratory test requests to the laboratory technician, sending medical requests to the pharmacy technician, accepting patient information from the receptionist, accepting laboratory reports from the laboratory technician, recording patient information, and sending the report to the patient. The health officer checks the situation of the patient and passes the patient information to the doctor. The system will also have a registration interface which can be accessed by the card receptionist by logging in using name and password.

The pharmacy technician will also have an account which requires the name of the pharmacy technician and password to enter the account. They will be able to receive medical requests from the doctor and dispense drugs to the patient. The laboratory technician will also have an account in this system. This account also requires a password and username like other accounts. The laboratory technician will be able to receive test requests from the doctor and send test results to the doctor. This system will make the work environment fast and easier.

2.2 Database Structure

Miza tepi University student's clinic database is designed to store information on patients, medical staff, and medical records. The database has different main tables some of the tables are patient medical card, includes the patient name, age, department, Id no, card no, gender, medical history are recorded. This is the central table in the student clinic database. Clinical prescription form consists the patient name, Card no, patient ID , age, and patient department, sex, medications, and the name of the prescriber(nurse) will be recorded. The prescription form is used to gather essential medical information from patients.

MTU University students clinics also consists variety of service rooms and medical professionals (health professionals) with various roles, such as pharmacist, medical laboratory technician, Nurses, students clinic Administrator. All the health professionals in this clinic are works only on his profession.Pharmacist have many responsibilities including, Dispensing medications prescribed by the healthcare provider to patients, providing medication counseling to patients, such as information on how to take medications, and drug interaction. Pharmacist also has their own ID card,name, specialization, personal phone number.

Pharmacist handles prescription giving form which is used to record patient name, patient ID, department , gender, pharmacist name, drug name.Drugs are the main items that stored in pharmacy room each drug have their identification number, name, quantity, expiry date, manufactured date, and other pharmaceutical also store Medical Laboratory Technician who are responsible for maintaining accurate records of laboratory test results and patient information medical laboratory technician also

has their own ID card, name, specialization, personal phone number.

Nurses are responsible for a wide range of tasks, including:- assessing patients health status, taking vital signs, and documenting patient history, administering medications, vaccines, assisting with diagnostic tests and procedures, such as blood draws. Nurse who work in this clinic have their own ID card, Name, specialization, personal phone number.

Administrator Office:- administrator office of MTU university students clinic is responsible for managing the overall operations and supervising staff members, including nurses, medical assistants, and pharmacist, developing and implementing policies and procedures for the clinic. For managing the health professionals in this clinic they are used their ID, full name, phone number, qualifications.

2.3 Conceptual Schema And Logical Schema Design Phase

Under conceptual phase there is some terminology

1. Entity : An entity in a database is a thing, place, person or object that is independent of another. In databases, you store information about things in the real world, and these things are database entities. Types:

A. Week Entity: represented by double rectangle.

B. Strong Entity: represented by single rectangle.

2. Attribute: Attribute is a piece of data that describes an entity. Types:

A. simple (automatic) vs. composite entity: Simple contains single value while Composite divide into sub parts.B. Single valued vs. multi valued attribute Single value: have single value and represented by oval shape. Multi value: have more than one shape and represented by double oval.

C. Stored vs. Divide attribute: Stored: not possible to divide. Divide: The value may be derived, represented by dot line oval.D. Null values: The attribute which are not applicable but the value and key attribute cannot be null.E. Default value: Assumed value if not exist value or unknown.

3. Relationship : a situation that exist between two relational database tables when one table has a foreign key that reference the primary key of the other table.

2.4 Identifying Entity

There are eight(8) entities in this database .

1.Patient.

2.Nurse.

3.Pharmacist

4.Drug

5.Medical lab technician

6.Clinical prescription form

7.Prescription record

8.Administrator office

2.5 Identifying Attribute

1.Patient

- Id(primary key)
- name
- age
- card no(foreign key)
- gender
- department
- medical history

2.Nurse

- Id(primary key)
- name
- phone no
- specialization

3.Pharmacist

- Id(primary key)
- name
- phone no
- specialization

4.Drug

- Id no(primary key)
- Name
- Quantity
- manufactured date
- Expired date

5.Medical lab technician

- id (primary key)
- phone no
- Name
- specialization

6.Clinical prescription form

- patient id (foreign key)
- patient name
- Card no(primary key)
- age
- sex
- patient department
- nurse name
- medications

7.Prescription record

- id no(foreign key)
- patient name
- patient id(foreign key)
- pharmacist name
- drug name
- Dept.
- gender

8.Administrator office

- staff id(foreign key)
- name
- qualification
- phone no

Entity Relationship Diagram

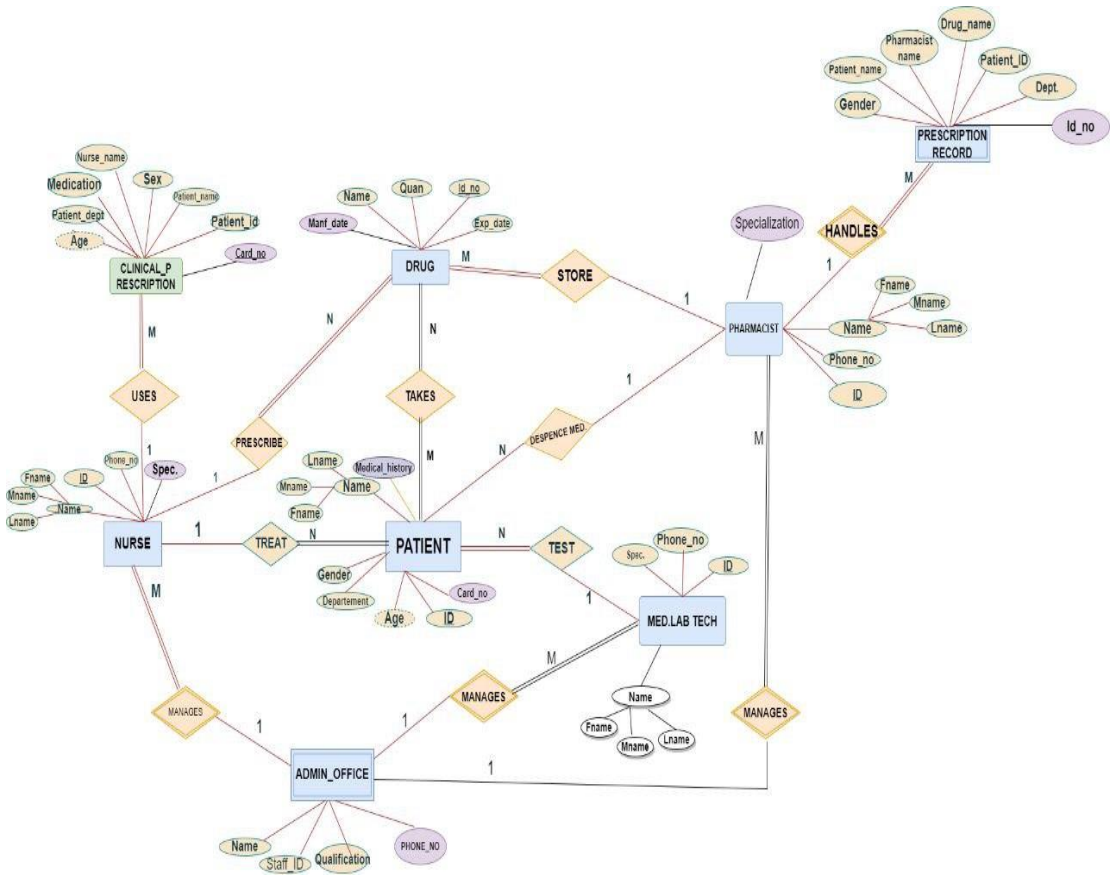


Figure 1: ER Diagram for Mtu clinical management system

2.6 Logical Design

Logical design is the next stage of designing a database system where the conceptual design is represented in a more formal and structured manner. In this stage the entities and relationships identified in the conceptual design are transformed into tables for the database. In the Mtu clinical Management system the logical design would involve creating tables for Patient, Nurse Pharmacist, Drug .Medical lab technician, .Clinical prescription form ,Prescription record and Administrator office Each table would have columns representing the attributes of the entity such as ID, name . Relationships between entities would be established by

using primary and foreign keys.

Table 1: patient Table

```
create database MTU_clinical_management_sytem;

use MTU_clinical_management_sytem;

create table patient(id varchar (20)primary key,

name varchar(30)not null,

age     int not null,

card_no    int,

gender    char(1)not null,

departement  varchar(30)not null,

medical_history varchar(50)not null);

insert into  patient(id,name,age,card_no,gender,departement,medical_history)

values(1251,'abera',33,1112,'M','cs','do not have history'),

          (1252,'azeb',20,1113,'F','cs','cancer'),

        (1555,'abgaz',24,4453,'M','agroeconomics','malaria'),  (1599,'amir',18,9908,'M','economics','tv virus'),

          (3344,'almaz',22,4677,'F','It','do not have history');

Select *from patient;
```

	id	name	age	card_no	gender	departement	medical_history
1	1251	abera	33	1112	M	cs	do not have history
2	1252	azeb	20	1113	F	cs	cancer
3	1555	abgaz	24	4453	M	agroeconomics	HIV virus
4	1599	amir	18	9908	M	economics	tv virus
5	3344	almaz	22	4677	F	It	do not have history

:Table 2 drug:

```
use MTU_clinical_management_sytem;
```

```
create table drug(
id_no      int primary key,
name varchar(30)not null,
quantity  varchar(30)not null,
manufactured_date date not null,
expired_date   date not null);
insert into  drug(id_no,name,quantity,manufactured_date,expired_date)
values(7,'amoxasiln','200g','2022-02-20','2022-10-10'),
(2,'insulin','250ml','2022-05-2','2022-08-26'),
(1,'parastamol','150g','2024-08-15','2026-2-16'),
(4,'amiliph','250','12-08-2021','2023-08-11'),
(3,'backlophan','500g','2024-07-23','2027-05-19');
```

Result table:

	id_no	name	quantity	manufactured_date	expired_date
1	1	parastamol	150g	2024-08-15	2026-02-16
2	2	insulin	250ml	2023-05-02	2022-08-26
3	3	backlophan	500g	2024-07-23	2027-05-19
4	4	amiliph	250	2021-12-08	2023-08-11
5	7	amoxasiln	200g	2022-02-20	2022-10-10

Table 3: nurse

```
use MTU_clinical_management_sytem;
create table nurse(
id       int primary key,
name varchar(30)not null,
phone_No  varchar(30)not null,
specialization varchar(30)not null);
insert into nurse(id,name,phone_No,specialization)
values(2541,'kemal abdella',+25936251467,'not specialized'),
```

(1254,'ramedan abdullahi',+2591023568479,'B.nurse'),

(2153,'abdullaziz umar',+25918693631,'B.nusre');

Result table:

	id	name	phone_No	specialization
1	1254	ramedan abdullahi	2591023568479	B.nurse
2	2153	abdullaziz umar	25918693631	B.nusre
3	2541	kemal abdella	25936251467	not specialized

Table 4 :clinical_pr

```
use MTU_clinical_management_sytem;

create table clinical_pr(

patient_id    varchar(20),

patient_name   varchar(30)not null,

card_no   int primary key,

age       int not null,

sex      char(1)not null,

patien_departement   varchar(30)not null,

nurse_name    varchar(20)not null,

madications  varchar(30)not null);

alter table clinical_pr add foreign key(patient_id)references patient(id);

insert into
clinical_pr(patient_id,patient_name,card_no,age,sex,patien_departement,nurse_name,madications)

values(1251,'abdulahi kemal',1254,23,'M','IS','kemal abdella','drops'),

(1252,'jamal umer',2351,31,'M','CS','remadan abdullahi','injections'),

          (1555,'kedir usman',2355,21,'M','agriculture','umar abdulaziz','capsules');
```

Result table:

	patient_id	patient_name	card_no	age	sex	patien_departement	nurse_name	madications
1	1251	abdulahi kemal	1254	23	M	IS	kemal abdella	drops
2	1252	jamal umer	2351	31	M	CS	remadan abdullahi	injections
3	1555	kedir usman	2355	21	M	agriculture	umar abdulaziz	capsules

Table 5:pharmacist

```
use MTU_clinical_management_system;

create table pharmacist(

id     int primary key,

name varchar(30)not null,

phone_no  varchar(30)not null,

specialization varchar(30)not null);

insert into  pharmacist(id,name,phone_no,specialization)

values (9457,'jemal isak',+259653485524,'pharmacy practices'),

                    (5234,'umar kemal',+25965412365,'quality assurances'),

                              (2156,'adulleziz',+25932651478,'pharmaceutics');
```

	id	name	phone_no	specialization
1	2156	adulleziz	25932651478	pharmaceutics
2	5234	umar kemal	25965412365	quality assurances
3	9457	jemal isak	259653485524	pharmacy practices

Table 6:laboratory

```
use MTU_clinical_management_sytem;

create table libratoy(

id       varchar(30)primary key,

name varchar(30)not null,

phone_no   varchar(30)not null,
```

```
specialization varchar(40)not null);

insert into libratoy(id,name,phone_no,specialization)

values(1273,'musa jemal',+25932561461,'clinical chemistry'),

(6954,'jibril ibrahim',+25945862434,'microbilogy'),

(2965,'nejmudin husen',+25965354125,'immunology');
```

	id	name	phone_no	specialization
1	1273	musa jemal	25932561461	clinical chemistry
2	2965	nejmudin husen	25965354125	immunology
3	6954	jibril ibrahim	25945862434	microbilogy

Table 7:admin_office

```
use MTU_clinical_management_sytem;

create table admiin_office(

stuff_id        int,

name varchar(20)not null,

phone_no       varchar(20)not null,

qualification    varchar(30)not null);


alter table admiin_office add foreign key(stuff_id)references pharmacist(id);


insert into admiin_office(stuff_id,name,phone_no,qualification)

values(9457,'jamal kedir',+25987465123,'effective'),

(5234,'ibrahim jamal',+25965321478,'multitasking'),

            (2156,'emran mohammed',+25962351485,'unique ability')
```

	stuff_id	name	phone_no	qualification
1	9457	jamal kedir	25987465123	effective
2			25965321478	multitasking
3	2156	emran mohammed	25962351485	unique ability

Click to select all grid cells

Table 8:prescription _record

```
use MTU_clinical_management_
sytem;

create table prscription_record(

id_no        int,

patient_name    varchar(30)not null,

patient_id       varchar(20),

pharmacist_name    varchar(30)not null,

drug_name        varchar(30)not null,

gender         char(1)not null,

departement       varchar(20)not null);

alter table prscription_record add foreign key (id_no)references drug(id_no);

alter table prscription_record add foreign key(patient_id)references patient(id);

insert into
prscription_record(id_no,patient_name,patient_id,pharmacist_name,drug_name,gender,departeme
nt)

values(7,'kemal jamal',1251,'umar jirbil','amoxasiln','M','agriculure'),

(2,'usman jibril',1252,'umar kemal','amiliph','M','IT'),

            (1,'hana ahmad',1555,'abdulaziz','parastamol','F','IS');
```

	id_no	patient_name	patient_id	pharmacist_name	drug_name	gender	departement
1	7	kemal jamal	1251	umar jirbil	amoxasiln	M	agriculure
2	2	usman jibril	1252	umar kemal	amiliph	M	IT
3	1	hana ahmad	1555	abdulaziz	parastamol	F	IS

2.7 Database Diagram

After drawing one table for each entity in MTU clinical management system now it'stime to draw database diagram. A database diagram is a diagram that links two ormore table based on the relationship they have, and this relationship is determinedby keys: Primary and foreign key. From the above eight tables we can see thatattributes of one table is also found in another table.

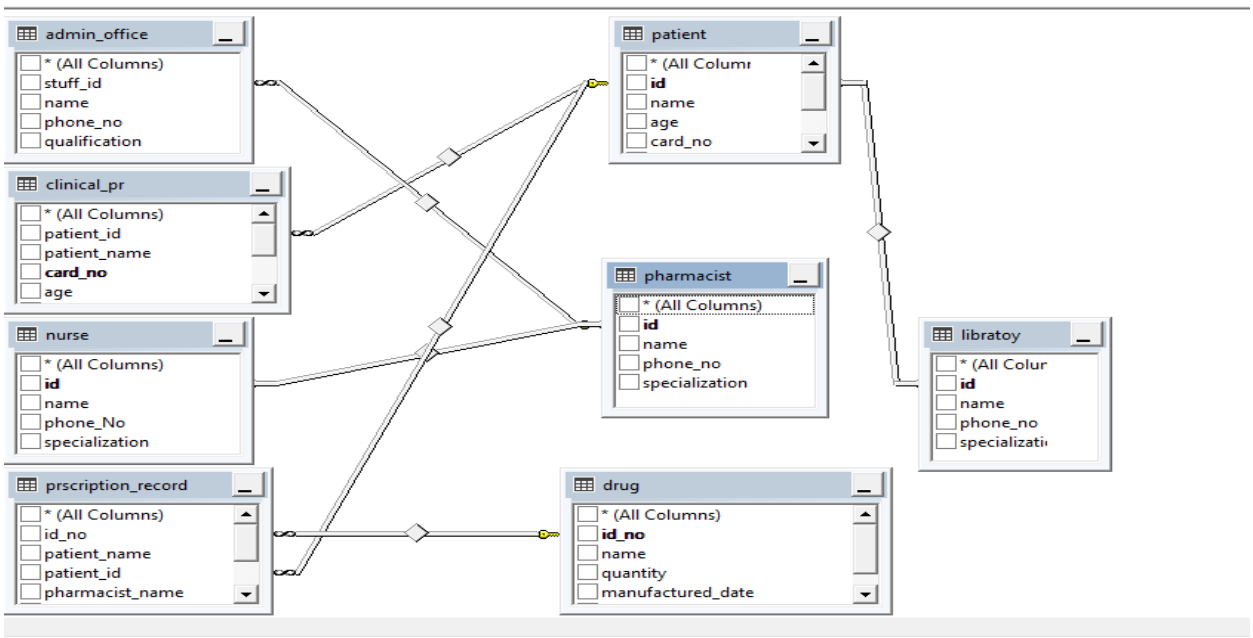


Figure : Database diagram for MTU clinical management systeem

2.8Normalization

For the table listed above we will try to get the normal form of a sample of two tables called patient .

Normalization for table patient

PATIENT table is already in the first normal form (1NF) since it has a unique primary key and each column contains atomic values.

Table 9 1st normal form of patient table

Id	Name	Age	Gender	Card_no	Department	Medical history
1251	Abera	33	M	1112	Cs	Dont have history
1252	Azebe	20	F	1113	Cs	Cancer
1255	Abegaz	24	M	4453	Agroeconomics	Hiv virus
1599	Amir	18	M	9908	Economic	Tv virus
3344	Almaz	22	F	4677	It	Dont have history

Second normal form for table patient

A table is in 2NF if it is in 1NF and every non-key attribute is fully dependent on the primary key. In the given TABLE, all attributes are dependent on the primary key, Card_No. Therefore, the table is already in 2NF.

Table 10 2nd normal form of patient table

Id	Name	Age	Gender	Card_no	Department	Medical history
1251	estifanos	33	M	1112	Cs	Dont have history
1252	Azebe	20	F	1113	Cs	Cancer
1255	Abegaz	24	M	4453	Agroeconomics	Hiv virus
1599	Amir	18	M	9908	Economic	Tv virus
3344	Almaz	22	F	4677	It	Dont have history

Third normal form for table patient

In 3NF,a table in 2NF and all transitive dependencies have been removed.This means that no non-prime attribute depends on another non-prime attribute.

Table 11 3rd normal form for table patient

Id	Name	Age	Gender	Card_no	Department
1251	Abera	33	M	1112	Cs
1252	Azebe	20	F	1113	Cs
1255	Abegaz	24	M	4453	Agroeconomics
1599	Amir	18	M	9908	Economic
3344	Almaz	22	F	4677	It

Table 12 3rd normal form for table patient

Card_no	Medical history
1112	Dont have history
1113	Cancer
4453	Hiv virus
9908	Tv virus
4677	Dont have history

CHAPTER THREE

3.1 Recommendation

Based on the findings of this project, it is recommended that Mtu University'sclinical management system be changed from a manual and file-based system to adatabase system. The benefits of a database system, including centralized datastorage, improved data access, and better data sharing, make it a more efficient andeffective solution for managing the clinic functions of the university.

The main recommendations based on the limited scope of this project are:

- The conceptual and logical designs proposed could be used as a starting point for implementing an actual clinical management system for MTU University, if decided.
- Further refinements to the design may be needed based on more detailed requirements gathering and use case analysis.
- Additional planning would be required to implement security features, data validation, interfaces, and other aspects of a full system.

In addition, the structured methodology used in this project, which involved defining the scope, identifying user requirements, choosing the right technology, designing the ER diagram, and designing the database diagram using SQL tables, can be applied to other similar projects in the future. This will help ensure that the clinical management system remains up-to-date, efficient, and effective, and that the university can continue to attract and retain the best talent.

3.2 Conclusion

In conclusion, the clinical management system at MTU University is in need of a major overhaul. The current manual and file-based system is time-consuming, inefficient, and prone to errors, and it does not provide the necessary reporting and analytics capabilities to support the strategic goals of the university. By changing to a database system, the university can simplify its clinical processes, reduce costs,

improve accuracy, and enhance its overall competitiveness.

The use of interviews and questionnaires helped to gather the necessary data and understand the needs of the users, while the feasibility analysis helped to determine the viability of the project. The methodology used in this project, which involved defining the scope, limitations, constraints, identifying user requirements, and designing the database schema, helped us to finish the project within the given resources including time.

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